

Brian Wonjun Lee

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EDUCATION

University of Pennsylvania

MSE in Computer Graphics & Game Technology (CGGT), Submatriculation

BA in Computer Science & Design (Double Major), Benjamin Franklin Scholars

Relevant Coursework: Interactive Computer Graphics, Computer Animation, 3-D Computer Modeling, Introduction to Algorithms, Introduction to Computer Systems, Programming Languages & Techniques

Philadelphia, PA

Expected May 2028

Expected May 2027

EXPERIENCE

Software Engineer Intern, Aleph Lab (Y Combinator)

June 2026 – Present · San Francisco (Remote)

- Shipped a production retention system that re-engages lapsed users with targeted push notifications (Supabase, Deno edge functions, FCM) to recover them as paying subscribers, instrumented with a **treatment/holdout A/B**
- Expanded reachable users **~2.7×** by validating the win-back targeting against live production data and correcting how eligible users were identified
- Turned the company's production AI agent into a drop-in package other teams can embed in their own apps — a typed SDK with versioned APIs and **136 automated tests**; a new AI coding agent integrated it into a fresh app on the **first attempt**
- Diagnosed and fixed a production authentication outage (a JWT signing-key migration) that had silently broken the notification pipeline, restoring it **from failing (401) to healthy (200)**

Software Engineer Intern, BitMango (Puzzle1 Studio)

June 2025 – Aug 2025 · Pangyo, South Korea

- Maintained a live portfolio of **50+ Unity mobile game titles**, improving UI/UX and core gameplay stability across releases
- Resolved **100+ QA-reported bugs** by modifying Unity prefabs and C# scripts, shipping fixes in production releases
- Upgraded SDKs and platform modules fleet-wide to meet evolving app-store requirements and sustain production availability

Project Lead & Product Designer, Penn Spark

Jan 2025 – Present · Philadelphia, PA

- Led a cross-functional team of **15+** designers and developers to ship client-facing **0-to-1 products**
- Defined end-to-end user flows, interaction patterns, and reusable interface systems across multiple deliverables
- Shipped polished, user-centered features across semester-long client project cycles

Military Interpreter, Capital Defense Command (ROK Army)

Dec 2022 – June 2024 · Seoul, South Korea

- Provided real-time English–Korean interpretation in high-security, multi-agency environments for international stakeholders
- Translated sensitive documents and briefings with precision, enabling coordination across government and defense teams

Database Engineer Intern, IT-Farm

June 2022 – Aug 2022 · Seongnam, South Korea

- Designed and implemented a SQL database prototype managing **1M+ data entities** using Oracle and SQLite
- Processed and organized semiconductor datasets from **20+ client companies**, including Samsung, SK, and LG
- Wrote and optimized SQL queries to improve retrieval and management efficiency across large-scale datasets

PROJECTS

Mini Minecraft | C++, OpenGL, GLSL

Apr 2026

- Engineered a **voxel engine from scratch**: 7 procedural biomes from layered Perlin/FBM/Voronoi noise with smoothstep blending
- Implemented **3D Perlin cave systems** with trilinear interpolation, fluid physics, and post-process tint overlays
- Built the render pipeline: **7×7 PCF soft shadows**, **screen-space reflections**, **vertex AO**, and a dynamic day-night cycle

Passenger | Unreal Engine 5, HLSL, Maya

Mar 2026 – Apr 2026

- Wrote 2 **HLSL post-process shaders** (anisotropic Kuwahara, cell shader) sampling GBuffer channels for stylized rendering
- Rigged and animated a character** in Maya (IK limbs, spline-IK spine) driving an Unreal Animation Blueprint
- Rendered at 2560×1080 via Movie Render Pipeline with **hardware ray tracing** over Lumen GI, 64 TSR samples/frame

Mini Maya | C++, OpenGL, Qt

Feb 2026 – Mar 2026

- Implemented a **half-edge mesh data structure** enabling efficient traversal and topology-aware editing operations
- Built edge split, extrusion, and **Catmull–Clark subdivision** on an OpenGL/VBO pipeline with GLSL shaders
- Built an interactive **Qt desktop editor** — vertex/edge/face selection, a GUI control panel, and a real-time OpenGL viewport

TECHNICAL SKILLS

Graphics & Rendering: OpenGL, GLSL, HLSL, shader programming, Unreal Engine 5, Lumen, real-time & offline rendering

3D & Tooling: Maya, Blender, Substance Painter, Qt, Movie Render Pipeline

Languages: C++, Python, TypeScript, JavaScript, Java, C#, GLSL/HLSL, SQL

Frameworks & Tools: React Three Fiber, Three.js, React, Node.js, Git, GitLab, VS Code